Scenario-Based Analysis Tasks

Total score 30 points

Participant ID: _____

Scenario 1: Conflict Identification in Ramp Merging (10 points)

Under medium-to-high traffic density conditions, a vehicle enters the mainline from a ramp. Mainline vehicles maintain relatively stable speeds, while the merging vehicle accelerates but fails to identify a suitable acceptable gap in time. At the end of the acceleration lane, the vehicle is forced to merge into the mainline, causing a sharp deceleration of the following mainline vehicle.

Q1. Identify the primary type of traffic conflict in this scenario. (2 points)

Q2. Where is the most critical location for the formation of this conflict? Briefly explain your reasoning. (3 points)

Q3. What are the main factors that contribute to this conflict? (3 points)

Q4. Propose one improvement measure from either a driving behavior or traffic management perspective. (2 points)

Scenario 2: Multi-Vehicle Interaction in a Weaving Section (10 points)

In a weaving section, a vehicle entering from an on-ramp intends to move toward the inner lane, while a mainline vehicle plans to exit downstream. At the same time, traffic density is relatively high, and the available space for lane changing is limited. As a result, both vehicles must complete their lane changes within a short distance, leading to path crossing and mutual interference.

Q1. Why is this scenario more complex than a standard ramp merging scenario? (2 points)

Q2. What vehicle interaction behaviors jointly contribute to the conflict risk in this scenario? (3 points)

Q3. How does high traffic density affect the development of conflicts in this scenario? (3 points)

Q4. From a traffic engineering perspective, what is the most critical influencing factor in this scenario? Briefly explain your choice. (2 points)

Scenario 3: Knowledge Transfer and Strategy Evaluation (10 points)

Students have learned the mechanisms of traffic conflict formation in ramp merging and weaving sections. Consider a new scenario in which the distance between an upstream entrance and a downstream exit is short. Entering vehicles and exiting vehicles perform lane-changing maneuvers simultaneously within a limited road segment. In addition, traffic speeds are unevenly distributed, and some drivers tend to perform lane changes only when approaching their target position.

Q1. Based on your prior knowledge, does this scenario exhibit conflict formation mechanisms similar to those in weaving sections? Explain your reasoning. (3 points)

Q2. What indicators suggest that the conflict risk may increase in this scenario? (3 points)

Q3. Compared with a standard ramp merging scenario, what are the key differences in the sources of conflict in this scenario? (2 points)

Q4. If this scenario were included in classroom teaching, what key instructional point should be emphasized? Briefly explain your choice. (2 points)

Scoring criteria

These criteria describe the original item scores. The separate written-response feature codes are documented in Written Response Coding and Review.

Scenario 1-1

2 points: Clearly identifies a ramp-merging conflict, a merging conflict, a conflict caused by forced merging, or a conflict between the merging vehicle and the following mainline vehicle.

1 point: Mentions a general vehicle conflict, merging risk, or lane-changing interference without clearly identifying a merging conflict.

0 points: Incorrect or irrelevant, for example a rear-end crash, a signal-control conflict, a pedestrian conflict, or a statement such as "it is dangerous" without identifying the conflict type.

Scenario 1-2

3 points: Identifies the end of the acceleration lane, the end of the merging area, the ramp–mainline junction where acceptable gaps are insufficient, or the location of forced merging. The answer identifies the merging-end or forced-merging location as the critical point.

2 points: Identifies a broadly correct area, such as the merging area, the vicinity of the ramp entrance, or the ramp–mainline junction, without sufficient precision.

1 point: Gives a vague location, such as "near the mainline," "the lane-changing location," or "the conflict area," without identifying forced merging at the end.

0 points: Gives an incorrect location or no answer, for example the exit ramp, an upstream free-flow section, or the service-area entrance.

Scenario 1-3

Acceptable scoring points:

Insufficient acceptable gaps or difficulty finding a suitable gap in time.

Inadequate speed coordination or a large speed difference between merging and mainline vehicles.

High mainline traffic density or flow reducing merging space.

Forced merging at the end of the acceleration lane or a late merging decision.

Insufficient evasive space for the following mainline vehicle, causing sharp deceleration.

3 points: Includes at least three key points with basically clear logic.

2 points: Includes two key points.

1 point: Includes one key point, or lists several vague factors such as "many vehicles" and "high speed."

0 points: Unrelated to the conflict mechanism or clearly incorrect.

Scenario 1-4

Acceptable examples:

Driving behavior: observe mainline traffic and select an acceptable gap earlier; adjust speed earlier to improve matching with mainline traffic; avoid last-moment forced merging; or have mainline vehicles yield appropriately and adjust headways in advance.

Traffic operation or engineering: extend the acceleration lane; improve markings and warning information; use speed limits or guidance to improve speed coordination; or apply ramp control during high-flow periods.

2 points: Proposes a clear, basically feasible measure directly addressing the core conflict cause.

1 point: Proposes a broadly relevant but general or insufficiently targeted measure, such as "drive more slowly," "strengthen management," or "improve safety awareness."

0 points: Irrelevant, unreasonable, or inconsistent with the scenario, for example stopping the merging vehicle to wait, encouraging mainline vehicles to accelerate without yielding, or removing the merging area.

Scenario 2-1

Key features:

Bidirectional lane-changing demand from entering and exiting traffic.

Competition among vehicles for paths.

Maneuvers concentrated within a short distance, creating spatial compression.

Interaction among multiple vehicles rather than one vehicle merging into the mainline.

2 points: Clearly identifies at least two key features.

1 point: States only that the scene is "more complex," or identifies one factor, such as "many vehicles" or "many lane changes," without explaining the fundamental difference.

0 points: Does not identify a characteristic of weaving or gives an incorrect answer.

Scenario 2-2

Key points, one point each up to three points:

The entry vehicle changes inward to join the mainline.

The mainline vehicle changes outward toward the exit.

The maneuvers overlap in time or space.

Vehicle paths cross or conflict.

Gaps are insufficient or maneuvering space is limited.

3 points: Identifies at least three key points with basically clear logic.

2 points: Identifies two key points.

1 point: Identifies one key point or gives a vague account, such as "everyone is changing lanes."

0 points: Does not identify the core interaction.

Scenario 2-3

Key points, one point each:

Available lane-changing space or gaps decrease.

Vehicle spacing decreases, leaving less tolerance for maneuvers.

Lane changing becomes more difficult and forced lane changes become more likely.

Conflicts can intensify, for example through sharp deceleration or increased interference.

The available maneuvering or decision window becomes shorter.

3 points: Identifies at least three key points.

2 points: Identifies two key points.

1 point: Identifies one key point or makes a general statement such as "more vehicles are more dangerous."

0 points: Irrelevant or incorrect.

Scenario 2-4

Acceptable core factors:

Insufficient weaving length.

High traffic density or flow.

High lane-changing demand and bidirectional interaction.

Spatial compression that intensifies path competition.

Speed differences and inadequate maneuver coordination.

2 points: Identifies one clear, reasonable factor that explains the conflict mechanism and fits the scenario.

1 point: Identifies a broadly relevant but vague factor, such as "poor traffic conditions" or "too many vehicles."

0 points: Incorrect or irrelevant.

Scenario 3-1

Key mechanism points:

Bidirectional lane-changing interactions, involving entry and exit, resemble a weaving section.

Vehicle paths compete or cross.

The short distance between entry and exit creates spatial compression.

Multiple vehicles complete maneuvers within a short distance, intensifying interaction.

3 points: Correctly identifies a similar mechanism and explains at least two key mechanism points, such as bidirectional lane changing and spatial compression.

2 points: Correctly identifies the similarity but explains one mechanism point or provides a limited explanation.

1 point: Gives only a verdict of similar or not similar without explaining the mechanism, or gives a vague explanation such as "because there are many vehicles."

0 points: Incorrect judgment or a completely irrelevant answer.

Scenario 3-2

Key points, one point each up to three points:

The entrance and exit are very close, limiting space.

Entering and exiting demands occur simultaneously, producing bidirectional lane changing.

Traffic speeds are unevenly distributed.

Drivers change lanes only near their target location.

Lane-changing or decision time is short.

Traffic density is high.

3 points: Identifies at least three key points.

2 points: Identifies two key points.

1 point: Identifies one key point or gives a vague description such as "it is dangerous" or "there are many vehicles."

0 points: Irrelevant or incorrect.

Scenario 3-3

Key differences:

Ramp merging primarily involves one-directional entry into the mainline.

The current scene involves simultaneous entry and exit, producing bidirectional lane changing.

The current scene has stronger path crossing or path competition.

Maneuvers are more concentrated within a shorter distance.

2 points: Identifies at least two fundamental differences, such as one-directional versus bidirectional interaction and path competition.

1 point: Identifies one difference or gives a vague explanation.

0 points: Does not identify a meaningful difference or gives an incorrect answer.

Scenario 3-4

Acceptable teaching priorities:

Conflict mechanisms in weaving or short weaving sections.

Path competition and multi-vehicle interaction.

The effect of spatial compression on conflict risk.

Risks of late lane-changing behavior.

Speed differences and maneuver coordination.

2 points: Specifies a clear teaching point directly related to the conflict mechanism and suitable for instruction, rather than a general statement.

1 point: Gives a broadly relevant but general priority, such as "pay attention to safety" or "improve awareness."

0 points: Irrelevant or unreasonable.